

Christopher (Christy) Bruce

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PROFESSIONAL EXPERIENCE

Imperial College London

Research Assistant (Quantitative Researcher)

London and Amsterdam

Feb 2026 - Present

- Developed a sustainability-linked innovation index scheduled for public listing in the Japanese market for institutional clients, working across the full chain from signal design to portfolio construction and performance evaluation.
- Generated factor signals from text-derived company initiatives, linking corporate actions, SDGs and stakeholders for sustainability-related equity index design.
- Co-designed a signal measuring corporate sustainability maturity that shows low correlation to third-party ESG scores.
- Built a modular Python research pipeline (pandas/Polars): content-addressed computation nodes, per-run manifests recording code version and data hashes at every step, parallel-run validation against the frozen reference implementation, and interactive audit and *portfolio-analytics dashboards*.
- Long-short SDG-linked factor split by materiality: 1.19 annualised Sharpe, 0.72% monthly FF3 alpha ($p < 0.01$).
- Co-authored a working paper and two research reports on the US and Japanese markets.
- Produced the analysis answering research questions from institutional clients and presented findings to non-technical directors in weekly meetings.
- Collaborated with senior researchers/post-docs, who were ex-BlackRock and Man AHL quants.
- Contributed to building the backtesting suite, minimising survivorship bias in the infrastructure.
- Scraped, matched and built alternative dataset pipelines, particularly related to ESG data.

All Options

Quant Trainee Programme (selective)

Amsterdam

Sep 2025 - Oct 2025

- Selected with a 0.5% acceptance rate to the programme, involving an intensive course in **option theory**, along with market-making games and hands-on shadowing of options traders and quantitative researchers.

Introspect Capital

Quantitative Analyst Intern

London

Jun 2024 - Sep 2024

- Backtested long-short market-neutral pairs-trading strategies.
- Developed, from my master's thesis on dynamic equity networks, a way of selecting equity pairs that improved stability of the cointegration p-value per pair by 20%, improving tradability and lowering drawdowns.
- Designed SQL and Python pipelines to synchronise and clean data for 500+ stocks, ensuring data integrity for backtesting and validating using pandas and Numpy.
- Leveraged XGBoost for predictive modelling of financial time-series, to predict the mean-reversion of spreads.
- Communicated technical findings in simple terms for senior management, in presentations and reports.

EDUCATION

Loughborough University

MSc Financial Mathematics | Distinction (80.5%, Course Rank: 2nd, Scholarship)

Loughborough, UK

Sep 2022 - Oct 2023

BSc Mathematics Major (Natural Sciences) | First-class honours

Sep 2019 - Jun 2022

Relevant Modules: Statistics for Large Data (Data Science, XGBoost, Random Forests) (83%),
Time-series Econometrics (95%), Economic Data Analysis (90%), Probability Theory (79%),
Financial Economics (76%) (asset pricing, portfolio optimisation), Derivative Pricing (73%)

Achievements: UKMT Mathematical Olympiad — Bronze award, Presented Thesis at the NetSci 2026 conference.

Thesis: Financial Complex Systems & Networks, observing dynamic correlation relationships over time (Scored 93%)

TECHNICAL SKILLS

Programming: Python (pandas, Polars, NumPy, statsmodels, scikit-learn, XGBoost), R, SQL, Git; LLM-assisted Development (Claude).

Asset Pricing: Factor construction, Fama-French attribution, quantile sorts, long-short portfolios, econometrics.

Data & Analytics: Compustat/WRDS, Fama-French factors, MSCI/Refinitiv/S&P ESG, SASB, GICS; backtesting, risk and performance metrics.

RESEARCH PROJECTS

Master's Thesis Presented at a Conference (Complex Systems & Networks) (Scored 93%)

Loughborough

Researcher

Jun 2023 - Present

- Built a Python algorithm tracking dynamic correlation structure across ~2,500 equities over 2 years, identifying how systemic shocks (e.g. Covid-19) altered clustering, risk concentration and market leadership — a basis for network-cluster approaches to portfolio construction.
- Presented at the NetSci conference to an audience of 50+, translating complex network methods into clear insights.

Custom Machine Learning Model from Scratch (Python)

Amsterdam

Researcher

Apr 2025 - Jan 2026

- Designed and implemented a probabilistic machine-learning architecture to forecast financial time-series, estimating expected values via conditional probabilities derived from historical binary pattern sequences.
- Built an independent out-of-sample backtesting engine with a dynamic volatility-scaling overlay using GARCH models and model-confidence metrics.
- Presented the methodology to the Amsterdam Investment Club, winning an award for creativity.

Prediction Market Systematic Model (Event-Driven) (Python)

Amsterdam

Co-Researcher

Oct 2025 - Jan 2026

- Engineered an event-driven statistical model to analyse event-contract markets, deriving implied probability distributions and evaluating deviation from market consensus.
- Developed automated API pipelines using Claude Code for real-time data ingestion, cleaning, and state-tracking across fragmented data sources.
- Backtested and deployed the model live, reaching a top-5% rank on Polymarket in 2025, modelling and exploiting mispriced probabilities.

IMC Prosperity, Algorithmic Trading Challenge

London

Co-Researcher

Apr 2025 - May 2025

- Placed in the top 3% out of 12,000+ global teams in a multi-round algorithmic simulation.

LEADERSHIP & EXTRACURRICULAR

Amsterdam Sahaj Yoga Meditation Course

Amsterdam

Co-Organiser

Oct 2025 - Feb 2026

- Led weekly meditation workshops and one-to-one sessions, presenting on balance and focus.

Quant Finance Research Meet-up

French Alps

Co-Organiser

Jan 2024 - Jun 2024

- Initiated and organised an informal quantitative-finance research gathering for 10 peers, presenting four of my own research projects.